

PRACTICAL : 7

Aim. Write R program on several built-in math functions on mtcars dataset.

Problem statement :

Consider mtcars data set and perform all the statistical functions on mtcars. Apply on any one variable of mtcars.

Theory :

Mathematical Computations Using R :

R Programming Language is widely used for statistical computing and data analysis. One of its core strengths lies in its ability to perform various mathematical computations efficiently. Here, we'll explore different methods and functions available in R for mathematical operations.

1. Basic Arithmetic Operations

R can handle basic arithmetic operations like addition, subtraction, multiplication, and division just like a calculator.

2. Mathematical Computations Using Base R Functions

The base R package provides numerous functions for mathematical computations. Now calculate the mean and standard deviation of a numeric vector.

3. Mathematical Computations Using apply() Functions

The apply() family of functions applies a function across the rows or columns of a matrix or data frame. So we calculate the sum of each row in a matrix.

Built-in Math Functions

R also has many built-in math functions that allow you to perform mathematical tasks on numbers.

For example, the min() and max() functions can be used to find the lowest or highest number in a set:

:

Code :

```
1 # Load mtcars dataset
2 data(mtcars)
3
4 # Display the first few rows
5 print("First few rows of mtcars:")
6 print(head(mtcars))
7
8 # Select one variable - mpg (miles per gallon)
9 mpg_data <- mtcars$mpg
10
11 # Display the selected variable
12 print("Selected variable: mpg")
13 print(mpg_data)
14
15 # Statistical functions on mpg
16
17 mean_mpg <- mean(mpg_data)
18 median_mpg <- median(mpg_data)
19 mode_mpg <- as.numeric(names(sort(table(mpg_data), decreasing=TRUE)[1])) # Mode
20
21 sd_mpg <- sd(mpg_data)
22 var_mpg <- var(mpg_data)
23 min_mpg <- min(mpg_data)
24 max_mpg <- max(mpg_data)
25 range_mpg <- range(mpg_data)
26 quantiles_mpg <- quantile(mpg_data)
27 summary_mpg <- summary(mpg_data)
28
29 # Display results
30
31 cat("\nStatistical Summary of mpg:\n")
32 cat("Mean:", mean_mpg, "\n")
33 cat("Median:", median_mpg, "\n")
34 cat("Mode:", mode_mpg, "\n")
35 cat("Standard Deviation:", sd_mpg, "\n")
36 cat("Variance:", var_mpg, "\n")
37 cat("Minimum:", min_mpg, "\n")
38 cat("Maximum:", max_mpg, "\n")
39 cat("Range:", range_mpg[1], "to", range_mpg[2], "\n")
40 cat("Quantiles:\n")
41 print(quantiles_mpg)
42
43 cat("Summary:\n")
44 print(summary_mpg)
45
```

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Output :

```

> # Load mtcars dataset
> data(mtcars)
>
> # Display the first few rows
> print("First few rows of mtcars:")
[1] "First few rows of mtcars:"
> print(head(mtcars))
      mpg   cyl  disp    hp  drat    wt   qsec    vs  am gear carb
Mazda RX4     21.0   6  160   110  3.90  2.620  16.46   0   1   4   4
Mazda RX4 Wag  21.0   6  160   110  3.90  2.875  17.02   0   1   4   4
Datsun 710     22.8   4  108   93  3.85  2.320  18.61   1   1   4   1
Hornet 4 Drive  21.4   6  258  110  3.08  3.215  19.44   1   0   3   1
Hornet Sportabout 18.7  8  360  175  3.15  3.440  17.02   0   0   3   2
Valiant        18.1   6  225  105  2.76  3.460  20.22   1   0   3   1
>
> # Select one variable - mpg (miles per gallon)
> mpg_data <- mtcars$mpg
>
> # Display the selected variable
> print("Selected variable: mpg")
[1] "Selected variable: mpg"
> print(mpg_data)
[1] 21.0 21.0 22.8 21.4 18.7 18.1 14.3 24.4 22.8 19.2 17.8 16.4 17.3 15.2 10.4 10.4 14.7 32.4 30.4 33.9 21.5 15.5 15.2 13.3 1
[27] 26.0 30.4 15.8 19.7 15.0 21.4
>
> # Statistical functions on mpg
>
> mean_mpg <- mean(mpg_data)
> median_mpg <- median(mpg_data)
> mode_mpg <- as.numeric(names(sort(table(mpg_data), decreasing=TRUE)[1])) # Mode
>
> sd_mpg <- sd(mpg_data)
> var_mpg <- var(mpg_data)
> min_mpg <- min(mpg_data)
> max_mpg <- max(mpg_data)
> range_mpg <- range(mpg_data)
> quantiles_mpg <- quantile(mpg_data)
> summary_mpg <- summary(mpg_data)
>
> # Display results
>
> # Display results
>
> cat("\nStatistical Summary of mpg:\n")

Statistical Summary of mpg:
> cat("Mean:", mean_mpg, "\n")
Mean: 20.09062
> cat("Median:", median_mpg, "\n")
Median: 19.2
> cat("Mode:", mode_mpg, "\n")
Mode: 10.4
> cat("Standard Deviation:", sd_mpg, "\n")
Standard Deviation: 6.026948
> cat("Variance:", var_mpg, "\n")
Variance: 36.3241
> cat("Minimum:", min_mpg, "\n")
Minimum: 10.4
> cat("Maximum:", max_mpg, "\n")
Maximum: 33.9
> cat("Range:", range_mpg[1], "to", range_mpg[2], "\n")
Range: 10.4 to 33.9
> cat("Quantiles:\n")
Quantiles:
> print(quantiles_mpg)
      0%    25%    50%    75%   100%
10.400 15.425 19.200 22.800 33.900
>
> cat("Summary:\n")
Summary:
> print(summary_mpg)
      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 10.40   15.43   19.20   20.09   22.80   33.90
>

```

Conclusion : R program on several built-in math functions on mtcars dataset is executed successfully .